

5. Sea anemones are animals which are often found in rock pools along the seashore. **Image 5.1** shows a sea anemone.

Image 5.1

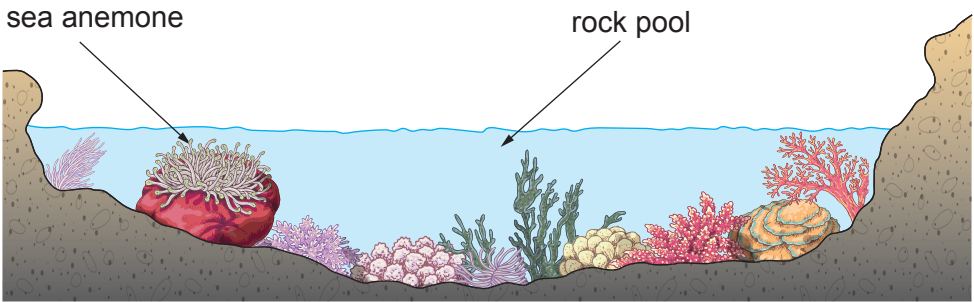


sea anemone

- On warm sunny days:
- the water temperature in rock pools increases.
 - evaporation occurs, so the concentration of salts in the water increases.
 - the water becomes acidic (pH lower than 7).
 - concentration of salt in sea water is **HIGHER** than anemone cells.
 - concentration of water in sea water is **LOWER** than anemone cells.

Image 5.2 shows a rock pool at the end of a warm day.

Image 5.2



Fact file – Conditions required for sea anemones to survive.

Oxygen concentration (mg / dm ³)	Above 9.9
pH	Between 8.1 and 9.7



Table 5.3 shows how the sea water temperature affects its oxygen concentration.

Table 5.3

Sea water temperature (°C)	Oxygen concentration (mg/dm ³)
0	14.5
10	11.2
20	9.4
30	7.6
40	4.5

Use the information to answer the following questions.

- (a)

(i)

State how the oxygen concentration of sea water changes when the temperature increases.

[1]
- (ii)

Calculate the mean change in oxygen concentration **per degree** when the temperature rises from 10°C to 30°C.

[2]

Mean change in oxygen concentration per degree = mg/dm³

- (b)

The statements in **Table 5.4** refer to conditions in a rock pool at the end of a warm, sunny day when the sea water temperature was 24°C. Write **true** or **false** for each statement.

[4]

Table 5.4

Statements	True or False
Water vapour has been lost from the rock pool.	
The sea anemone has lost salt by diffusion.	
There is enough oxygen in the water for the sea anemone to survive.	
The pH of the water is suitable for the sea anemone.	
Water will pass out of the sea anemone by osmosis.	

